

Final Endeavor for the Generalized t -SNE Problem

Yi Gu

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1 Progress So Far and Setup

OK this is it. We have reached this step and the end is within our grasp. The stake is high and I can literally feel the adrenaline coursing my vein, along with my fear. So far we have finished two crucial steps. First, we have finished the proof of convergence for the generalized output kernel. Denote $\mathbf{Ker}_\mathbf{O}$ the output kernel and $\mathbf{Ker}_\mathbf{I}$ the input kernel. The main theorems in [1] will hold if the following is true:

Condition 1 (Output Kernel Conditions). *The output kernel $\mathbf{Ker}_\mathbf{O}(y, y') = k(\|y - y'\|)$ depends only on the distance between y and y' and that:*

1. $k : \mathbb{R}^+ \rightarrow \mathbb{R}^+$ is decreasing and bounded;
2. g satisfies

$$\iint_{(0, \infty)^2} g(y, y') dy dy' < \infty; \quad (1.1)$$

3. k has bounded derivative;
4. $k'(0) = 0$.

Moreover, the lemmas that deal with the output does not interfere with those related to the input kernel. The initial exploration we did around the output kernel not only provides us a good opportunity to understand the problem and the literature, but also serves as an integral part of the final paper. If we are able to tackle the lemmas related to the input kernel we will be able to join the results seamlessly.

On the other hand, we have established the form of the generalized input kernel, set a basis for us to work on, and prove the existence of $\sigma_{\rho, \mu}^*(x)$, the unique solution to $F_{\rho, \mu}(x, \sigma) = 0$. For this to happen, the generalized input kernel should satisfy the following conditions:

Condition 2 (Initial Input Kernel Conditions). *Let $w : \mathbb{R}_{\geq 0} \rightarrow \mathbb{R}_{\geq 0}$ be some fixed function. The input kernel has the form of $\mathbf{Ker}_\mathbf{I}(x_i, x_j, \sigma_i) = \exp(-w(\sigma_i \|x_j - x_i\|^\theta))$. Here, σ_i is used to control entropy (thus perplexity) and $\theta > 0$ is a fixed constant. (We can replace σ_i with $\psi : \mathbb{R}^d \rightarrow \mathbb{R}^+$ in the continuous version of the kernel.) w satisfies:*

1. $w'(t) > 0$ for all $t \geq 0$ with $\lim_{t \rightarrow \infty} w(t) = \infty$;
2. $w'(t) + tw''(t) \geq 0$ for all $t \geq 0$;
3. The following two integrals converge:

$$\int_0^\infty t^{d-1} \exp(-w(t^\theta)) dt < \infty, \quad \int_0^\infty t^{d-1} w(t^\theta) \exp(-w(t^\theta)) dt < \infty. \quad (1.2)$$

Since the probability masses are all normalized, we can assume without loss of generality that $w(0) = 0$. Note that condition 2 is only enough to prove the existence and uniqueness of σ^* . We may add more assumptions later on if necessary. Before we march towards the final battle, let us remind ourselves some important definitions and notations, as well as some of the lemmas from [1] that we can borrow. Recall that X or the corresponding probability measure μ is called sub-Gaussian if $\mathbb{P}(X > t) \leq 2 \exp(-Ct^2)$ for some positive constant C .

Lemma 1. *Let ν be sub-Gaussian with second moment equal to 1 and suppose $X \in \mathbb{R}^d$ has i.i.d. entries drawn from ν . Then there exists constant C, c_1 such that*

$$\mathbb{P}(\|X\| \geq \sqrt{d} + t) \leq Ce^{-c_1 t^2}. \quad (1.3)$$

Lemma 2. *Let μ be sub-Gaussian and c_1 be the constant from lemma 1. Let $X_1, \dots, X_n \in \mathbb{R}^d$ have i.i.d. entries drawn from μ . Then there exists a constant C such that*

$$\mathbb{P}\left(\max_{i=1, \dots, n} \|X_i\| \geq \sqrt{\frac{\log n}{c_1}} + \sqrt{d} + t\right) \leq Ce^{-c_1 t^2}. \quad (1.4)$$

Lemma 3. *Let $\delta > 0$ be constant, $K_n = [\delta, \sqrt{n}]$ and $h : \mathbb{R}^d \times \mathbb{R}^d \times \mathbb{R} \rightarrow \mathbb{R}$. Suppose the following hold:*

1. μ is sub-Gaussian;
2. $h(x, x', \sigma)$ and $\nabla_x h(x, x', \sigma)$ are bounded for all x, x' and all $\delta \in K_n$;
3. $h(x, x', \sigma) \rightarrow 0$ as $\sigma \|x - x'\| \rightarrow \infty$.

Let $X_1, \dots, X_n$ be i.i.d. random variables drawn from μ . Let $\mu_n = (1/n) \sum_i \delta_{X_i}$. Then for any $\epsilon > 0$ with probability at least $1 - e^{-c\epsilon^2 n}$ we have

$$\sup_{\sigma^{-1} \in K_n, x} \left| \int h(x, x', \sigma) d\mu_n(x) - \int h(x, x', \sigma) d\mu(x') \right| \leq \epsilon + O\left(\sqrt{\frac{\log n}{n}}\right). \quad (1.5)$$

Finally, let us also recall that

$$\begin{aligned} & F_{\rho, \mu}(x, \sigma) \\ &= \int \frac{\exp(-w(\sigma \|x - x'\|^\theta))}{\int \exp(-w(\sigma \|x - x'\|^\theta)) d\mu(x')} \log \left(\frac{\exp(-w(\sigma \|x - x'\|^\theta))}{\int \exp(-w(\sigma \|x - x'\|^\theta)) d\mu(x')} \right) d\mu(x') + \log \rho \\ &= - \int \frac{w(\sigma \|x - x'\|^\theta) \exp(-w(\sigma \|x - x'\|^\theta))}{\int \exp(-w(\sigma \|x - x'\|^\theta)) d\mu(x')} d\mu(x') \\ &\quad - \log \left(\int \exp(-w(\sigma \|x - x'\|^\theta)) d\mu(x') \right) + \log \rho. \end{aligned} \quad (1.6)$$

The two terms in the formula of F are the pillars of this manuscript. Their convergence will lead us to prove the convergence of F , and most importantly, the uniform convergence of $\sigma(x)$. [Yi: As for the question Tuca asked me during our discussion, yes we can define $\sigma(x)$ to be the unique solution to $F(x, \sigma) = 3$ or whatever number we peek as long as the number is greatly than zero (to ensure the existence of uniqueness of the solution). If we move the constant to the left it is equivalent to just pick a different perplexity. The convergence of F works for any σ and that is the key. As long as control the entropy, we are expecting convergence as long as we set up the continuous version to have the same perplexity.]

2 The Two Most Important Lemmas

The biggest obstacles left are to proof two lemmas similar to lemma 6 and lemma 7 in [1]. These two lemmas will give us the convergence of F and thus the convergence of $\sigma(x)$. We

will try to tackle this part using similar methods from the original paper. Write

$$\begin{aligned} f_\sigma(x, x') &= \exp(-w(\sigma \|x - x'\|^\theta)), \\ g_\sigma(x, x') &= w(\sigma \|x - x'\|^\theta) \exp(-w(\sigma \|x - x'\|^\theta)). \end{aligned} \quad (2.1)$$

Note that for any real numbers A, B, C, D , we have

$$\left| \frac{A}{B} - \frac{C}{D} \right| = \left| \frac{AD - BC}{BD} \right| = \left| \frac{D(A - C) + C(D - B)}{BD} \right| \leq \left| \frac{A - C}{B} \right| + \left| \frac{C(D - B)}{BD} \right|. \quad (2.2)$$

This means

$$\begin{aligned} \left| \frac{\int g_\sigma(x, x') d\mu_n(x')}{\int f_\sigma(x, x') d\mu_n(x')} - \frac{\int g_\sigma(x, x') d\mu(x')}{\int f_\sigma(x, x') d\mu(x')} \right| &\leq \left| \frac{\int g_\sigma(x, x') d\mu_n(x') - \int g_\sigma(x, x') d\mu(x')}{\int f_\sigma(x, x') d\mu_n(x')} \right| \\ &+ \left| \frac{\int g_\sigma(x, x') d\mu(x') (\int f_\sigma(x, x') d\mu_n(x') - \int f_\sigma(x, x') d\mu(x'))}{\int f_\sigma(x, x') d\mu(x') \int f_\sigma(x, x') d\mu_n(x')} \right|. \end{aligned} \quad (2.3)$$

Let $\alpha > 0$. Define

$$E'_1 = \left\{ \max_{i=1, \dots, n} \|X_i\| \geq \sqrt{\left(\alpha + \frac{1}{c_1}\right) \sqrt{\log n} + \sqrt{d}} \right\}. \quad (2.4)$$

Setting $t = \sqrt{\alpha \log n}$ in lemma 2 we have $\mathbb{P}(E'_1) \leq Cn^{-c_1\alpha}$. To absorb the $\sqrt{d}$ term in E'_1 , we can find $\epsilon_n \rightarrow 0$ such that

$$E_1 = \left\{ \max_{i=1, \dots, n} \|X_i\| \geq \sqrt{\left(\alpha + \frac{1}{c_1} + \epsilon_n\right) \sqrt{\log n}} \right\} \quad (2.5)$$

with $\mathbb{P}(E_1) \leq Cn^{-c_1\alpha}$. On the other hand, on E_1^c for all i, j ,

$$\|X_i - X_j\| \leq 2\sqrt{\left(\alpha + \frac{1}{c_1} + \epsilon_n\right) \sqrt{\log n}} =: 2\gamma\sqrt{\log n}. \quad (2.6)$$

This means for $\sigma^{-1} \in [\delta^\theta, n^{\theta/2}]$ on E_1^c we have

$$\begin{aligned} \int f_\sigma(X_i, x') d\mu_n(x') &= \int \exp(-w(\sigma \|X_i - x'\|^\theta)) d\mu_n(x') \\ &\geq \exp\left(-w\left(\left(\frac{2\gamma\sqrt{\log n}}{\delta}\right)^\theta\right)\right) =: T. \end{aligned} \quad (2.7)$$

[Yi: This is where the interesting part begins. The proofs in the original paper shows us that we do not want our kernel to decay too fast otherwise the bounds cannot be controlled.]In a same way, by lemma 1, we get similar bounds

$$\int f_\sigma(X_i, x') d\mu(x') \geq (1 - Cm^{-c_1\alpha})T \geq \frac{1}{2}T. \quad (2.8)$$

By lemma 3 with $h = f_\sigma$ and $h = g_\sigma$, there exists E_2 with $\mathbb{P}(E_2) \leq 2e^{-c\epsilon^2 n}$ such that on E_2^c ,

$$\begin{aligned} \sup_{\substack{\sigma^{-1} \in [\delta^\theta, n^{\theta/2}] \\ i=1, \dots, n}} \left| \int f_\sigma(X_i, x') d\mu_n(x') - \int f_\sigma(X_i, x') d\mu(x') \right| \\ + \sup_{\substack{\sigma^{-1} \in [\delta^\theta, n^{\theta/2}] \\ i=1, \dots, n}} \left| \int g_\sigma(X_i, x') d\mu_n(x') - \int g_\sigma(X_i, x') d\mu(x') \right| \leq \epsilon + O\left(\sqrt{\frac{\log n}{n}}\right). \end{aligned} \quad (2.9)$$

Therefore on $E_1^c \cap E_2^c$, combining (2.3), (2.7), (2.8), and (2.9) gives us

$$\sup_{\substack{\sigma^{-1} \in [\delta^\theta, n^{\theta/2}] \\ i=1, \dots, n}} \left| \frac{\int g_\sigma(x, x') d\mu_n(x')}{\int f_\sigma(x, x') d\mu_n(x')} - \frac{\int g_\sigma(x, x') d\mu(x')}{\int f_\sigma(x, x') d\mu(x')} \right| \leq C\epsilon T^{-1} + O\left(\frac{\sqrt{\log n}}{n^{1/2}T}\right). \quad (2.10)$$

Again, we have reduced the core parts of this problem into a calculus exercise. We want RHS goes to zero as $n \rightarrow \infty$. Towards this end we need

$$\epsilon T^{-1} \rightarrow 0, \quad T^{-1} \cdot \frac{\sqrt{\log n}}{n^{1/2}} \rightarrow 0. \quad (2.11)$$

On the other hand, since $\mathbb{P}(E_1 \cap E_2) \leq Cn^{-c_1\alpha} + 2e^{-c\epsilon^2 n}$, we also need

$$c_1\alpha > 1, \quad \epsilon^2 n \rightarrow \infty. \quad (2.12)$$

Let $\epsilon = n^{-\eta}$ and by (2.11) we have $0 < \eta < 1/2$. Then, if we have $n^{-\eta}T^{-1} \rightarrow 0$ for some $\eta \in (0, 1/2)$, then we must have

$$T^{-1} \cdot \frac{\sqrt{\log n}}{n^{1/2}} \leq \frac{\sqrt{\log n}}{n^{1/2-\eta}} \rightarrow 0. \quad (2.13)$$

This means the first inequality in (2.11) implies the second. The requirements in (2.11) and (2.12) are equivalent to

$$n^{-\eta}T^{-1} \rightarrow 0, \quad c_1\alpha > 1, \quad 0 < \eta < 1/2. \quad (2.14)$$

i.e.,

$$n^{-\eta} \exp\left(w \left(\left(\frac{2\gamma\sqrt{\log n}}{\delta}\right)^\theta\right)\right) \rightarrow 0, \quad c_1\alpha > 1, \quad 0 < \eta < 1/2. \quad (2.15)$$

Let $\alpha = 1/c_1 + \epsilon_1$ and $\eta = 1/2 - \epsilon_2$. Then the above requirements are equivalent to

$$\lim_{n \rightarrow \infty} n^{-1/2+\epsilon_2} \exp\left(w \left(\left(\sqrt{\frac{8}{(c_1 + \epsilon_1)\delta^2}} \sqrt{\log n}\right)^\theta\right)\right) = 0. \quad (2.16)$$

As ϵ_1 and ϵ_2 can be arbitrarily small, the above limit can be further simplified to

$$\lim_{n \rightarrow \infty} n^{-1/2} \exp\left(w \left(\left(\sqrt{\frac{8}{c_1\delta^2}} \sqrt{\log n}\right)^\theta\right)\right) = 0. \quad (2.17)$$

Like [1], the quantity $c_1\delta^2$ shows up again as our slider. It is just less explicit. Also, we need to choose w that does not increase too fast that the above limit will fail. To put it in a more formal way, consider the set below

$$\mathcal{K}_w = \left\{ \tilde{C} \in \mathbb{R}_{\geq 0} \mid \lim_{n \rightarrow \infty} n^{-1/2} \exp\left(w \left(\left(\tilde{C}\sqrt{\log n}\right)^\theta\right)\right) = 0 \right\}. \quad (2.18)$$

Obviously, $0 \in \mathcal{K}_w$ so it is nonempty. If there exists $\tilde{C}' \in \mathcal{K}_w$ such that $\tilde{C}' > 0$. Then for any $0 \leq \tilde{C}'' < \tilde{C}'$, we have $\tilde{C}'' \in \mathcal{K}_w$ by monotonicity of w . Inspired by this fact, we can require

$$\sup \mathcal{K}_w > 0. \quad (2.19)$$

Then for any $\tilde{C} < \sup \mathcal{K}_w$, $\tilde{C} \in \mathcal{K}_w$. Back to (2.17), we simply need to let

$$c_1\delta^2 > \frac{8}{(\sup \mathcal{K}_w)^2}. \quad (2.20)$$

Now we are ready to formally state the two MOST important lemmas for this entire project.

Condition 3. Define

$$\mathcal{K}_w = \left\{ \tilde{C} \in \mathbb{R}_{\geq 0} \mid \lim_{n \rightarrow \infty} n^{-1/2} \exp \left(w \left(\left(\tilde{C} \sqrt{\log n} \right)^\theta \right) \right) = 0 \right\}. \quad (2.21)$$

Then w satisfies that

$$\sup \mathcal{K}_w > 0. \quad (2.22)$$

Lemma 4. Let c_1 be the decay constant of the sub-Gaussian distribution μ , and $X_1, \dots, X_n$ be i.i.d. random variables drawn from μ . Then almost surely in $X_1, \dots, X_n$,

$$\begin{aligned} \sup_{\substack{\sigma^{-1} \in [\delta^\theta, n^{\theta/2}] \\ i=1, \dots, n}} \left| \frac{\int g_\sigma(X_i, x') d\mu_n(x')}{\int f_\sigma(X_i, x') d\mu_n(x')} - \frac{\int g_\sigma(X_i, x') d\mu(x')}{\int f_\sigma(X_i, x') d\mu(x')} \right| \\ = O \left(n^{-1/2} \exp \left(w \left(\left(\sqrt{\frac{8}{c_1 \delta^2}} \sqrt{\log n} \right)^\theta \right) \right) \right). \end{aligned} \quad (2.23)$$

Using the elementary inequality $\log(x) \leq x - 1$, we can prove the next lemma in a similar way:

Lemma 5. Let c_1 be the decay constant of the sub-Gaussian distribution μ , and $X_1, \dots, X_n$ be i.i.d. random variables drawn from μ . Then almost surely in $X_1, \dots, X_n$,

$$\begin{aligned} \sup_{\substack{\sigma^{-1} \in [\delta^\theta, n^{\theta/2}] \\ i=1, \dots, n}} \left| \log \left(\int f_\sigma(X_i, x') d\mu_n(x') \right) - \log \left(\int f_\sigma(X_i, x') d\mu(x') \right) \right| \\ = O \left(n^{-1/2} \exp \left(w \left(\left(\sqrt{\frac{4}{c_1 \delta^2}} \sqrt{\log n} \right)^\theta \right) \right) \right). \end{aligned} \quad (2.24)$$

Then using these two lemmas we have the following two theorems:

Proposition 1. Let μ be sub-Gaussian with decay constant c_1 from lemma 1 and $X_1, \dots, X_n$ be i.i.d. random variables drawn from μ . $\mu_n = (1/n) \sum_i \delta_{X_i}$ is the empirical measure. Set $\delta > 0$ to be a constant such that $c_1 \delta^2 > 8/(\sup \mathcal{K}_w)^2$. Then almost surely in $X_1, \dots, X_n$,

$$\sup_{\substack{\sigma^{-1} \in [\delta^\theta, n^{\theta/2}] \\ i=1, \dots, n}} |F_{\mu_n}(X_i, \sigma) - F_\mu(X_i, \sigma)| = o(1). \quad (2.25)$$

Since in [1] the convergence of σ^* does not use the polynomial decay in the original paper, we have the following milestone proposition as well:

Proposition 2. Suppose μ has compact support and C_1 density. Then almost surely, as $n \rightarrow \infty$,

$$\sup_{i=1, \dots, n} |\sigma_{\mu_n}^*(X_i) - \sigma_\mu^*(X_i)| \rightarrow 0. \quad (2.26)$$

3 The Rest of the Proof

It is not over yet, as in [1] some of the remaining lemmas require polynomial decay and we need to thread carefully and tweak condition 3 to make sure the rest of the proof goes through, along with the generalized output kernels. We will cite some lemmas from [1] like we did for the output kernel computation so that the flow of the manuscript is easier to understand.

Lemma 6. *Any sequence in $\tilde{\mathcal{P}}_X$ is tight.*

Recall Prokhorov's theorem:

Theorem (Prokhorov's Theorem). *If μ_n is a tight sequence of measures in Euclidean space. There exists a subsequence μ_{n_k} and a measure μ such that μ_{n_k} converges to μ weakly.*

We immediately get the following lemma:

Lemma 7. *There exists μ^* such that*

$$I_\rho(\mu^*) = \inf_{\mu \in \tilde{\mathcal{P}}_X} I_\rho(\mu) \quad (3.1)$$

Recall that the lemmas concerning generalized output kernels do not interfere with the lemmas concerning the input kernel. We have a few more lemmas to check before victory. For simplicity of writing, notations we use without definitions will be same as in [1]. We start by looking at

$$\sup_{i,j} |p'_n(X_i, X_j) - p'(X_i, X_j)|. \quad (3.2)$$

Using triangular inequality, we have

$$\begin{aligned} |p'_n - p'| &= \left| \frac{f_n}{\int f_n d\mu_n - 1/n} - \frac{f}{\int f d\mu} \right| \\ &\leq \left| \frac{f_n}{\int f_n d\mu_n - 1/n} - \frac{f_n}{\int f_n d\mu_n} \right| + \left| \frac{f_n}{\int f_n d\mu_n} - \frac{f_n}{\int f d\mu_n} \right| \\ &\quad + \left| \frac{f_n}{\int f d\mu_n} - \frac{f_n}{\int f d\mu} \right| + \left| \frac{f_n}{\int f d\mu} - \frac{f}{\int f d\mu} \right|. \end{aligned} \quad (3.3)$$

In order to get the convergence, we need to check the convergence of each term. Note that $f, f_n \leq 1$. Denote

$$\tilde{T} = \exp \left(w \left(\left(\sqrt{\frac{8}{c_1 \delta^2}} \sqrt{\log n} \right)^\theta \right) \right). \quad (3.4)$$

The first can be bounded by using (2.7) and (2.17):

$$\left| \frac{f_n}{\int f_n d\mu_n - 1/n} - \frac{f_n}{\int f_n d\mu_n} \right| \leq \frac{1}{n} \left| \frac{1}{\int f_n d\mu_n - 1/n} \cdot \frac{1}{\int f_n d\mu_n} \right| = O(n^{-1} \tilde{T}^2). \quad (3.5)$$

Recall the definitions of $\mathcal{K}_w$ and $\tilde{T}$, we have $O(n^{-1} \tilde{T}^2) = o(1)$ if $c_1 \delta^2 > 8/(\sup \mathcal{K}_w)^2$. The second term is where we might need to tweak things a bit.

$$\left| \frac{f_n}{\int f_n d\mu_n} - \frac{f_n}{\int f d\mu_n} \right| \leq \frac{|f_n - f|}{|\int f_n d\mu_n| |\int f d\mu_n|} = O(|f_n - f| \tilde{T}^2). \quad (3.6)$$

We will postpone the estimation of $|f_n - f|$ as it will appear later in the lemmas we need to check. Continue onward, for the third term we can use lemma 3 and repeat the procedure that leads us to (2.10), which gives us

$$\left| \frac{f_n}{\int f d\mu_n} - \frac{f_n}{\int f d\mu} \right| \leq \frac{|\int f d\mu_n - \int f d\mu|}{|\int f d\mu| |\int f d\mu_n|} = O(n^{-1/2} \tilde{T}^2). \quad (3.7)$$

For the final term, we have a similar inequality as the second term:

$$\left| \frac{f_n}{\int f d\mu} - \frac{f}{\int f d\mu} \right| = O(|f_n - f| \tilde{T}). \quad (3.8)$$

By definition of $\tilde{T}$ and the monotonicity of w , if (3.6) vanishes, so does (3.8). On the other hand, clearly (3.7) needs a slightly stronger bound than what we already have from $\tilde{K}_w$. We will keep this in mind for the near future and there is not much left to check!

Next up, we want to bound

$$\sup_{i,j} |p_n(X_i, X_j) \log p_n(X_i, X_j) - p(X_i, X_j) \log p(X_i, X_j)|. \quad (3.9)$$

Let ϵ_n be such that

$$\frac{|f_n - f|}{|\int f_n d\mu_n| |\int f d\mu_n|} = O(|f_n - f| \tilde{T}^2) = O(\epsilon_n), \quad (3.10)$$

which we inherit from (3.6). We compute

$$\begin{aligned} p_n \log p_n &= p \log p + p \log \left(\frac{p_n}{p} \right) + (p_n - p) \log p_n \\ &= p \log p + p \log \left(\frac{p + O(\epsilon_n)}{p} \right) + O(\epsilon_n) \log p_n. \end{aligned} \quad (3.11)$$

The third term will apparently vanish. For the second term, recall that $p \leq \tilde{T}^{-1}$. We require

$$O(\epsilon_n)/p = O(\epsilon_n \tilde{T}) = o(1). \quad (3.12)$$

(3.12) also asks a little more in terms of convergence speed. We have one more lemma left to check before we put everything together. Lastly, we want to find the convergence condition for

$$\sum_j p_{j|i} \log p_{j|i} + \log(\rho n) = o(1). \quad (3.13)$$

Recall that $p_{j|i} = \frac{1}{n} p'_n(X_i, X_j)$. Therefore

$$\begin{aligned} \sum_j p_{j|i} \log p_{j|i} &= \sum_{j \neq i} \frac{1}{n} p'_n(X_i, X_j) \log \left(\frac{1}{n} p'_n(X_i, X_j) \right) + \log n + \log \rho \\ &= \frac{1}{n} \sum_{j \neq i} p'_n(X_i, X_j) \log p'_n(X_i, X_j) + \log \rho + \log n \left(1 - \frac{1}{n} \sum_{j \neq i} p'_n(X_i, X_j) \right) \\ &= \frac{1}{n} \sum_{j \neq i} p'_n(X_i, X_j) \log p'_n(X_i, X_j) + \log \rho \end{aligned} \quad (3.14)$$

because by definition $n - \sum_{j \neq i} p'_n(X_i, X_j) = 0$. We proceed to write the first summation as an integral over the measure μ_n , with the difference being the diagonal terms.

$$\begin{aligned} &\frac{1}{n} \sum_{j \neq i} p'_n(X_i, X_j) \log p'_n(X_i, X_j) - \int p'_n(X_i, x') \log p'_n(X_i, x') d\mu_n(x') \\ &= -\frac{1}{n} \frac{1}{\int f(X_i, x') d\mu_n(x') - \frac{1}{n}} \log \left(\frac{1}{\int f(X_i, x') d\mu_n(x') - \frac{1}{n}} \right). \end{aligned} \quad (3.15)$$

If μ_X has compact support, then the above diagonal term is $O(n^{-1})$. If μ_X is sub-Gaussian, then by (2.7), (3) and condition 2, we get

$$\int f(X_i, x') d\mu_n(x') - \frac{1}{n} \geq \frac{1}{2} \tilde{T}^{-1}. \quad (3.16)$$

This means the diagonal term in (3.15) is $O(n^{-1}\tilde{T}\log(\tilde{T})) = o(1)$ by condition 2 and the definition of $\mathcal{K}_w$. The rest of convergence follows in the exact same way in [1].

At this point, we need to tweak condition 3 such that both

$$O(|f_n - f|\tilde{T}^3), O(n^{-1/2}\tilde{T}^2) \quad (3.17)$$

are $o(1)$. The second bound is easy to tweak, we simply need to tweak the definition of $\mathcal{K}_w$ to be

$$\mathcal{K}_w = \left\{ \tilde{C} \in \mathbb{R}_{\geq 0} \left| \lim_{n \rightarrow \infty} n^{-1/2} \exp \left(2w \left(\left(\tilde{C} \sqrt{\log n} \right)^\theta \right) \right) = 0 \right. \right\}. \quad (3.18)$$

Only one thing left! Let us take a look at $|f_n - f|$. By definition and assume $\sigma(x) \geq \sigma_n(x)$ WLOG (or we can always compute the other way around), we have

$$\begin{aligned} |f_n - f| &= \left| \exp(w(\sigma(x) \|x - x\|^\theta) - w(\sigma_n(x) \|x - x\|^\theta)) - 1 \right| \exp(w(-\sigma(x) \|x - x\|^\theta)) \\ &\leq \left| \exp(w(\sigma(x) \|x - x\|^\theta) - w(\sigma_n(x) \|x - x\|^\theta)) - 1 \right| \\ &= \left| \exp(w'(\tilde{\sigma}(x) \|x - x\|^\theta)(\sigma(x) - \sigma_n(x) \|x - x\|^\theta)) - 1 \right|. \end{aligned} \quad (3.19)$$

We will have to assume the derivative of w is bounded from above by some constant $\tilde{M}$. Also, if μ_X is sub-Gaussian, let C be the constant of decay for $X_i - X_j$ and we have

$$\mathbb{P}(\|X_i - X_j\|^2 \geq \log n) \leq 2 \exp(-\log n/C^2) = 2 \exp\left(\frac{1}{c^2}\right) \cdot \frac{1}{n} \rightarrow 0. \quad (3.20)$$

This means $\|X_i - X_j\|^2 = O(\log n)$ almost surely. Finally, we know that

$$\sigma(x) - \sigma_n(x) = O(F_{\rho,\mu}(x, \sigma) - F_{\rho,\mu}(x, \sigma_n)) = O(n^{-1/2}\tilde{T}). \quad (3.21)$$

Applying (3.21) and by the simple limit $\lim_{x \rightarrow 0}(e^x - 1)/x = 1$, the difference (3.19) can be bounded as

$$|f_n - f| = O(n^{-1/2}(\log n)^\theta \tilde{T}). \quad (3.22)$$

And we in turn obtain

$$O(|f_n - f|\tilde{T}^3) = O(n^{-1/2}(\log n)^\theta \tilde{T}^4), \quad (3.23)$$

which is stronger than the condition in 3.18 and such fact compells us to tweak the definition of $\mathcal{K}_w$ one last time to

$$\mathcal{K}_w = \left\{ \tilde{C} \in \mathbb{R}_{\geq 0} \left| \lim_{n \rightarrow \infty} n^{-1/2}(\log n)^\theta \exp \left(4w \left(\left(\tilde{C} \sqrt{\log n} \right)^\theta \right) \right) = 0 \right. \right\}. \quad (3.24)$$

And we are DONE.

4 Wrap Up and Conclusion

Except for coding and simulation, the problem of generalized t-SNE has been solved. The assumptions of output kernels stay unchanged as in condtion 1. We are now ready to state the conditions we need for input kernels.

Condition 4 (Final Input Kernel Conditions). *Let $w : \mathbb{R}_{\geq 0} \rightarrow \mathbb{R}_{\geq 0}$ be some fixed function. The input kernel has the form of $\mathbf{Ker}_I(x_i, x_j, \sigma_i) = \exp(-w(\sigma_i \|x_j - x_i\|^\theta))$. Here, σ_i is used to control entropy (thus perplexity) and $\theta > 0$ is a fixed constant. (We can replace σ_i with $\psi : \mathbb{R}^d \rightarrow \mathbb{R}^+$ in the continuous version of the kernel.) Define*

$$\mathcal{K}_w = \left\{ \tilde{C} \in \mathbb{R}_{\geq 0} \left| \lim_{n \rightarrow \infty} n^{-1/2} (\log n)^\theta \exp \left(4w \left(\left(\tilde{C} \sqrt{\log n} \right)^\theta \right) \right) = 0 \right. \right\}. \quad (4.1)$$

Then w satisfies that w satisfies:

1. $\forall t \geq 0, w'(t) > 0$ and $\exists M > 0$ s.t. $w'(t) \leq M$. $\lim_{t \rightarrow \infty} w(t) = \infty$;
2. $w'(t) + tw''(t) \geq 0$ for all $t \geq 0$;
3. The following two integrals converge:

$$\int_0^\infty t^{d-1} \exp(-w(t^\theta)) dt < \infty, \quad \int_0^\infty t^{d-1} w(t^\theta) \exp(-w(t^\theta)) dt < \infty; \quad (4.2)$$

4. $\sup \mathcal{K}_w > 0$.

References

- [1] Antonio Auffinger and Daniel Fletcher. Equilibrium distributions for t-distributed stochastic neighbour embedding, 2023, 2304.03727. URL <https://arxiv.org/abs/2304.03727>.
- [2] Aditya Ravuri and Neil D. Lawrence. Towards one model for classical dimensionality reduction: A probabilistic perspective on umap and t-sne, 2024, 2405.17412. URL <https://arxiv.org/abs/2405.17412>.
- [3] Pavlin G. Poličar and Blaž Zupan. Visualizing high-dimensional temporal data using direction-aware t-sne, 2024, 2403.19040. URL <https://arxiv.org/abs/2403.19040>.
- [4] Laurens van der Maaten and Geoffrey Hinton. Visualizing data using t-sne. *Journal of Machine Learning Research*, 9(2579-2605), 2008.
- [5] Stéphane Boucheron, Gábor Lugosi, and Pascal Massart. *Concentration Inequalities: A Nonasymptotic Theory of Independence*. Oxford University Press, 2013.
- [6] Ben Weinkove. Dimension reduction and the gradient flow of relative entropy, 2024, 2409.16963. URL <https://arxiv.org/abs/2409.16963>.